To: Editor
Journal of Chemical Information and Modeling

Subject: Submission of Manuscript: "AutoAntibiotic: A Validated Computational Pipeline for Structure-Based Virtual Screening Against MRSA PBP2a"

Dear Editor,

We submit for consideration our manuscript entitled "AutoAntibiotic: A Validated Computational Pipeline for Structure-Based Virtual Screening Against MRSA PBP2a" for publication in the Journal of Chemical Information and Modeling.

This work presents the AutoAntibiotic Discovery Pipeline v5.6.0, a fully automated, reproducible computational framework for structure-based virtual screening against MRSA PBP2a. The pipeline performs multi-conformer ensemble docking across three PBP2a conformers using AutoDock Vina with mechanism-restricted selectivity profiling against human serine hydrolases, MMFF94-based rescoring, and OpenMM complex minimisation and short MD for top candidates. The pipeline additionally supports flexible side-chain docking and an MD stability filter.

Key findings include:

- A validated protocol with core RMSD of 1.251 Å (Validated badge)
- AUC = 0.852, $EF_{1\%}$ = 8.14 for enrichment validation
- Two lead compounds with SI $\geq$ 2.0 (Strong tier): ALL_QU04 (primary lead, SI = 2.07, MMFF94 strain = 366 a.u., SA = 1.78) and BRICS_0022 (secondary, SI = 2.13, strain = 738 a.u.), both engaging all three catalytic residues (Ser403, Lys406, Tyr446)
- Two Promising-tier sulfonamide scaffolds ALL_SU08 (SI = 1.91, SA = 1.81) and ALL_SP03 (SI = 1.93, SA = 3.27) with confirmed catalytic-triad contacts
- OpenMM gas-phase minimisation confirmed local minima for all top candidates; short MD revealed significant ligand drift (RMSD 5.2–7.7 Å) for all candidates, and explicit-solvent MD (100 ps NPT) showed complete dissociation for the three candidates validated (RMSD 9.0–9.8 Å), requiring these compounds to be treated as scaffold hypotheses pending longer MD and/or induced-fit docking

The manuscript describes a rigorous, reproducible methodology with transparent validation criteria and a candid discussion of limitations, including the sensitivity of enrichment benchmarks to methodological choices. We believe this work will be of interest to the journal's readership working on antibiotic discovery and structure-based drug design.

All authors have approved the manuscript and agree with its submission to your journal.

Sincerely,

The AutoAntibiotic Agent
autoantibiotic@example.com